

Cheat Sheet: Integrating Visual and Video Modalities

Package/Method	Description	Code Example
Base64 response format	Instead of returning URLs, you can get images as base64 data for immediate use without downloading from a URL. Useful when you need to process or store the images directly.	<pre>import base64 from PIL import Image import io response = client.images.generate(model="dall-e-2", prompt="a white siamese cat", size="512x512", response_format="b64_json", # Get base64 instead of URL n=1,) // Convert base64 to image image_data = base64.b64decode(response.data[0].b64_json) image = Image.open(io.BytesIO(image_data)) image.show() # Display the image</pre>
Credentials setup	Sets up the credentials for accessing the watsonx API. The api_key is not needed in the lab environment, and the project_id is preset.	<pre>from ibm_watsonx_ai import Credentials import os credentials = Credentials(url="https://us-south.ml.cloud.ibm.com",) project_id="skills-network"</pre>
DALL-E 2 image generation	Uses DALL-E 2 to generate an image based on a text prompt. DALL-E 2 supports generations, edits, and variations, simultaneously allowing up to 10 images.	<pre>response = client.images.generate(model="dall-e-2", prompt="a white siamese cat", size="1024x1024", quality="standard", n=1,) url = response.data[0].url display.Image(url=url, width=512)</pre>

DALL-E 3 image generation	Uses DALL-E 3 to generate higher quality images. DALL-E 3 only supports image generation (no edits or variations) but produces more detailed, accurate images.	<pre>response = client.images.generate(model="dall-e-3", prompt="a white siamese cat", size="1024x1024", quality="standard", n=1,) url = response.data[0].url display.Image(url=url, width=512)</pre>
Effective prompting	Tips for crafting effective prompts to get better results from DALL-E models: • Be specific and detailed in your descriptions • Include artistic style references • Specify lighting, perspective, and composition • Add context or setting information	<pre>// Basic prompt prompt = "a cat" // Improved detailed prompt prompt = "a fluffy white siamese cat with blue eyes sitting on a window sill, golden hour lighting, soft shadows, shallow depth of field, professional photography style" // Artistic style prompt prompt = "a white siamese cat in the style of a Renaissance oil painting, dramatic lighting, rich colors, detailed fur texture"</pre>
File download	Function to download an image file from a URL if it doesn't already exist locally.	<pre>import requests def load_file(filename, url): # Download file if it doesn't already exist if not os.path.isfile(filename): print("Downloading file") response = requests.get(url, stream=True) if response.status_code == 200: with open(filename, 'wb') as f: f.write(response.content) else: print("Failed to download file. Status code:", response.status_code) else: print("File already exists")</pre>
Image captioning	Loop through the images to see the text descriptions produced by the model in response to the query, "Describe the photo".	<pre>user_query = "Describe the photo" for i in range(len(encoded_images)): image = encoded_images[i] response = generate_model_response(image, user_query) // Print the response with a formatted description print(f"Description for image {i + 1}: {response}/n/n")</pre>

Image display	Displays an image in the notebook using IPython's display functionality.	<pre> from IPython.display import Image Image(filename=filename_tin, width=300) </pre>
Image encoding	Encodes an image to base64 format for inclusion in the model request. This is necessary because JSON is text-based and doesn't support binary data directly.	<pre> import base64 import requests def encode_images_to_base64(image_urls): encoded_images = [] for url in image_urls: response = requests.get(url) if response.status_code == 200: encoded_image = base64.b64encode(response.content).decode("utf-8") encoded_images.append(encoded_image) print(type(encoded_image)) else: print(f"Warning: Failed to fetch image from {url} (Status code: {response.status_code})") encoded_images.append(None) return encoded_images </pre>
Message formatting	Creates a structured message containing both text and image data to send to the model.	<pre> messages = [{ "role": "user", "content": [{ "type": "text", "text": question }, { "type": "image_url", "image_url": { "url": "data:image/jpeg;base64," + encoded_string, } }] }] return messages </pre>
Model invocation	Sends the formatted message to the model and receives a response with an analysis of the image.	<pre> response = model.chat(messages=my_message_1) print(response["choices"][0]["message"]["content"]) </pre>
Model initialization	Initializes the vision model with specific	<pre> from ibm_watsonx_ai.foundation_models.schema import TextChatParameters from ibm_watsonx_ai.foundation_models import ModelInference model_id = 'ibm/granite-vision-3-2-2b' </pre>

	parameters for text generation.	<pre> params = TextChatParameters(temperature=0.2, top_p=0.5,) model = ModelInference(model_id=model_id, credentials=credentials, project_id=project_id, params=params) </pre>
Multiple images (DALL-E 2)	Generate multiple images at once with DALL-E 2 using the 'n' parameter. DALL-E 2 can generate up to 10 images in a single request.	<pre> response = client.images.generate(model="dall-e-2", prompt="a white siamese cat", size="1024x1024", quality="standard", n=4, # Generate 4 different images) // Access all generated images for i, image_data in enumerate(response.data): print(f"URL for image {i+1}: {image_data.url}") display.Image(url=image_data.url, width=256) </pre>
OpenAI client initialization	Creates an instance of the OpenAI client to interact with the API.	<pre> from openai import OpenAI from IPython import display client = OpenAI() </pre>
Object detection	Ask the model to define objects from a specific image.	<pre> image = encoded_images[1] user_query = "How many cars are in this image?" print("User Query: ", user_query) print("Model Response: ", generate_model_response(image, user_query)) </pre>
pip install	Installs the necessary Python libraries required for working with watsonx and vision models.	<pre>%pip install ibm-watsonx-ai==1.1.20 image==1.5.33 requests==2.32.0</pre>

Quality options	Quality settings for generated images: - DALL-E 2: Only supports "standard" - DALL-E 3: Supports "standard" (default) and "hd" for enhanced detail	<pre>// DALL-E 3 with high-definition quality response = client.images.generate(model="dall-e-3", prompt="a mountain landscape", size="1024x1024", quality="hd", n=1,)</pre>
Saving generated images	Save the generated images to your local filesystem for later use.	<pre>import requests // Save from URL response = client.images.generate(model="dall-e-2", prompt="a white siamese cat", size="1024x1024",) url = response.data[0].url image_data = requests.get(url).content with open("generated_cat.jpg", "wb") as f: f.write(image_data) print("Image saved to generated_cat.jpg")</pre>
Size options	Different size options available for DALL-E models: - DALL-E 2: 256x256, 512x512, 1024x1024 - DALL-E 3: 1024x1024, 1024x1792, 1792x1024	<pre>// DALL-E 2 with smaller size response = client.images.generate(model="dall-e-2", prompt="a white siamese cat", size="512x512", quality="standard", n=1,) // DALL-E 3 with widescreen format response = client.images.generate(model="dall-e-3", prompt="a beautiful landscape", size="1792x1024", quality="standard", n=1,)</pre>

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